Experiment - 2

Aim:

Convert the ER diagram into a relational schema. Write complete MySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

Relational Schema :-

CUSTOMER(
    CUSTOMER_ID PK,
    Name,
    Email UNIQUE,
    Password,
    DOB
)

CUSTOMER_PHONE(
    CUSTOMER_ID PK, FK,
    Phone PK
)

CUSTOMER_ALT_PHONE(
    CUSTOMER_ID PK, FK,
    AltPhone PK
)

SELLER(
    SellerID PK,
    SellerName,
    Email UNIQUE,
    Phone,
    GSTIN UNIQUE,
    AccountNo,
    BankName,
    IfscCode
)

CATEGORY(
    CategoryID PK,
    CategoryName,
    Description,
    ParentCategoryID FK
)

PRODUCT (
ProductID PK,
ProductName,
Description,
Price,
StockQty,
Weight,
SellerID FK,
CategoryID FK
)

PRODUCT_IMAGE (
ProductID PK, FK,
Image PK
)

ELECTRONICS (
ProductID PK, FK,
Brand,
Warranty,
PowerRating
)

CLOTHING (
ProductID PK, FK,
Size,
Colour,
Material,
)

BOOKS (
ProductID PK, FK,
ISBN UNIQUE,
Author,
Publisher.
)

ORDERS (
    OrderID PK,
    Customer_ID FK,
    OrderDate,
    TotalAmount,
    OrderStatus
)

ORDER_COUPONS (
    OrderID PK, FK,
    CouponCode PK
)

ORDER_ITEM (
    OrderID PK, FK,
    ItemNo PK,
    ProductID FK,
    Quantity,
    ItemPrice,
    Discount,
    TaxAmount
)

PAYMENT (
    PaymentID PK,
    OrderID FK,
    TransactionID UNIQUE,
    PaymentDate,
    Amount,
    PaymentMethod
)

DELIVERY(
DeliveryID PK,
OrderID FK,
AddressID FK,
TrackingID UNIQUE,
DeliveryStatus,
DeliveryDate,
ShippingCharge
)

ADDRESS(
AddressID PK,
CustomerID FK,
DoorNo,
Area,
City,
State,
Pincode,
AddressType
)

ADDRESS_MOBILE(
AddressID PK, FK,
MobileNo PK
)

SQL Queries

```
CREATE DATABASE ECOMMERCE;
USE ECOMMERCE;

CREATE TABLE CUSTOMER (
    CustomerID INT PRIMARY KEY,
    Name VARCHAR(50) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Password VARCHAR(100) NOT NULL,
    DateOfBirth DATE NOT NULL
);

CREATE TABLE CUSTOMER_PHONE (
    CustomerID INT NOT NULL,
    Phone VARCHAR(15) NOT NULL,
    PRIMARY KEY (CustomerID, Phone),
    FOREIGN KEY (CustomerID)
        REFERENCES CUSTOMER (CustomerID)
        ON DELETE CASCADE
);

CREATE TABLE CUSTOMER_ALT_PHONE (
    CustomerID INT NOT NULL,
    AltPhone VARCHAR(15) NOT NULL,
    PRIMARY KEY (CustomerID, AltPhone),
    FOREIGN KEY (CustomerID)
        REFERENCES CUSTOMER (CustomerID)
        ON DELETE CASCADE
);
```

```
CREATE TABLE SELLER(
    SellerID INT PRIMARY KEY,
    SellerName VARCHAR(50) NOT NULL,
    Email VARCHAR(100) UNIQUE NOT NULL,
    GSTIN VARCHAR(20) UNIQUE NOT NULL,
    AccountNO VARCHAR(15),
    IFSCcode VARCHAR(20),
    BankName VARCHAR(50)
);

CREATE TABLE SELLER_PHONEC
    SellerID INT NOT NULL,
    Phone VARCHAR(15) NOTNULL,
    PRIMARY KEY (SellerID,Phone),
    FOREIGN KEY (SellerID)
        REFERENCES SELLER(SellerID)
        ON DELETE CASCADE
);

CREATE TABLE CATEGRORY(
    CategoryID INT PRIMARYKEY,
    CategoryName VARCHAR(50) NOT NULL,
    Description VARCHAR(100),
    ParentCategoryID INT,
    FOREIGN KEY(ParentCategoryID)
        REFERENCES CATEGRORY(CategoryID)
        ON DELETE SET NULL
);
```

```
CREATE TABLE PRODUCT(
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(50) NOT NULL,
    Description VARCHAR(100),
    Price DECIMAL(10,2) NOT NULL,
    StockQty INT NOT NULL,
    Weight DECIMAL(8,2),
    SellerID INT NOT NULL,
    CategoryID INT,
    FOREIGN KEY(SellerID)
        REFERENCES SELLER(SellerID)
        ON DELETE CASCADE,
    FOREIGN KEY(CategoryID)
        REFERENCES CATEGORY(CategoryID)
        ON DELETE SET NULL
);

CREATE TABLE PRODUCT_IMAGE(
    ProductID INT NOT NULL,
    Image VARCHAR(200) NOT NULL,
    PRIMARY KEY(ProductID, Image),
    FOREIGN KEY(ProductID)
        REFERENCES PRODUCT(ProductID)
        ON DELETE CASCADE
);

CREATE TABLE ELECTRONICS(
    ProductID INT PRIMARY KEY,
    Brand VARCHAR(50),
    Warranty VARCHAR(30),
    PowerRating VARCHAR(30),
    FOREIGN KEY(ProductID)
        REFERENCES PRODUCT(ProductID)
        ON DELETE CASCADE
);
```

```
CREATE TABLE CLOTHING(
    ProductID INT PRIMARY KEY,
    Size VARCHAR(20),
    Colour VARCHAR(20),
    Material VARCHAR(30),
    FOREIGN KEY (ProductID)
      REFERENCES PRODUCT(ProductID)
      ON DELETE CASCADE
);

CREATE TABLE BOOKS(
    ProductID INT PRIMARY KEY,
    ISBN VARCHAR(20) UNIQUE,
    Author VARCHAR(50),
    Publisher VARCHAR(50),
    FOREIGN KEY(ProductID)
      REFERENCES PRODUCT(ProductID)
      ON DELETE CASCADE
);

CREATE TABLE ADDRESS(
    AddressID INT PRIMARY KEY,
    DoorNO VARCHAR(10),
    Street VARCHAR(50),
    Area VARCHAR(50),
    City VARCHAR(30),
    State VARCHAR(30),
    Pincode VARCHAR(10) NOT NULL,
    AddressType VARCHAR(20)
);
```

```
CREATE TABLE ADDRESS_MOBILE (
    AddressID INT NOT NULL,
    MobileNo VARCHAR(15) NOT NULL,
    PRIMARY KEY (AddressID, MobileNo),
    FOREIGN KEY (AddressID)
        REFERENCES ADDRESS(AddressID)
        ON DELETE CASCADE
);

CREATE TABLE ORDERS(
    OrderID INT PRIMARY KEY,
    OrderDate DATE NOT NULL,
    OrderStatus VARCHAR(20) NOT NULL,
    TotalAmount DECIMAL(10,2) NOT NULL,
    CustomerID INT NOT NULL,
    AddressID INT,
    FOREIGN KEY (CustomerID)
        REFERENCES CUSTOMER(CustomerID)
        ON DELETE CASCADE,
    FOREIGN KEY(AddressID)
        REFERENCES ADDRESS(AddressID)
        ON DELETE SET NULL
);

CREATE TABLE ORDER_COUPON(
    OrderID INT NOT NULL,
    CouponCode VARCHAR(30) NOT NULL,
    PRIMARY KEY(OrderID, CouponCode),
    FOREIGN KEY (OrderID)
        REFERENCES ORDERS(OrderID)
        ON DELETE CASCADE

);
```

```
CREATE TABLE ORDERITEM(
    OrderID INT NOTNULL,
    ItemNo INT NOTNULL,
    ProductID INT NOT NULL,
    ItemPrice DECIMAL(10,2) NOTNULL,
    Quantity INT NOT NULL,
    Discount DECIMAL(10,2),
    TaxAmount DECIMAL(10,2),
    PRIMARY KEY(OrderID, ItemNo),
    FOREIGN KEY(OrderID)
        REFERENCES ORDERS(OrderID)
        ON DELETE CASCADE,
    FOREIGN KEY(ProductID)
        REFERENCES PRODUCT(ProductID)
        ON DELETE CASCADE
);
CREATE TABLE PAYMENT(
    PaymentID INT PRIMARY KEY,
    PaymentDate DATE NOT NULL,
    Amount DECIMAL(10,2) NOT NULL,
    PaymentMethode VARCHAR(30) NOT NULL,
    TransactionID VARCHAR(30) UNIQUE,
    PaymentStatus VARCHAR(30) NOT NULL,
    OrderID INT UNIQUE NOT NULL,
    FOREIGN KEY(OrderID)
        REFERENCES ORDERS(OrderID)
        ON DELETE CASCADE
);
```

```
CREATE TABLE DELIVERY (
  DeliveryID INT PRIMARY KEY,
  DeliveryDate DATE,
  DeliveryStatus VARCHAR(20) NOT NULL,
  TrackingID VARCHAR(30) UNIQUE,
  DeliveryPartner VARCHAR(50),
  ShippingCharge DECIMAL(8,2)
  OrderID INT NOT NULL,
  AddressID INT,
  FOREIGN KEY(OrderID)
      REFERENCES ORDER(OrderID)
      ON DELETE CASCADE,
  FOREIGN KEY (AddressID)
      REFERENCES ADDRESS (AddressID)
      ON DELETE SET NULL
);
```

Sample Data

```
INSERT INTO CUSTOMER VALUES
(1, 'Shatakshi', 'shani@gmail.com', 'Shani27', '2007-02-27');
INSERT INTO CUSTOMER_PHONE VALUES
(1, '8707850897127');
INSERT INTO CUSTOMER_ALT_PHONE VALUES
(1, '9208271582217');
INSERT INTO SELLER VALUES
(101, 'TechStore', 'tech@gmail.com', 'GST123', '82345',
  'SBIN001', 'SBI');
INSERT INTO SELLER_PHONE VALUES
(101, '7007677439 24');
```

```
INSERT INTO CATEGORY VALUES
(201, 'Electronics', 'Electronic items', NULL);
INSERT INTO CATEGORY VALUES
(202, 'Mobiles', 'Mobile phones', 201);
INSERT INTO PRODUCT VALUES
(301, 'laptop', 'Gaming laptop', 75000, 10, 2.5, 101, 201);
INSERT INTO ELECTRONICS VALUES
(301, 'ASUS', '2 Years', '180W');
INSERT INTO PRODUCT_IMAGE VALUES
(301, 'laptop.jpg');
INSERT INTO ADDRESS VALUES
(401, '12', 'Eldeco', 'Transport Nagar',
'Kanpur', 'Uttar Pradesh', '160071', 'Home');
INSERT INTO ADDRESS_MOBILE VALUE
(401, '9876542310');
INSERT INTO ORDERS VALUES
(501, '2026-08-29', 'Placed', 95000, 1, 401);
INSERT INTO ORDERITEM VALUES
(501, 1, 301, 75000, 1, 1000, 9000);
INSERT INTO ORDER_COUPON VALUES
(501, 'WELCOME10');
INSERT INTO PAYMENT VALUES
(601, '2026-09-21', 74000, 'UPI',
'TXGN45', 'Paid', 501);
INSERT INTO DELIVERY VALUES
(701, NULL, 'Processing', 'TRK123', 'Delivery', 100, 501, 401);
```

Output :-

Output-1 :- SHOW TABLES

SHOW TABLES;

Tables_in_Ecommerce
ADDRESS
ADDRESS_MOBILE
BOOKS
CATEGORY
CLOTHING
CUSTOMER
CUSTOMER_ALT_PHONE
CUSTOMER_PHONE
DELIVERY
ELECTRONICS
ORDERS
ORDERITEM
ORDER_COUPON
PAYMENT
PRODUCT
PRODUCT_IMAGE
SELLER
SELLER_PHONE

Output-2 :- Sample data

SELECT * FROM CUSTOMER;

SELECT * FROM PRODUCT;

SELECT * FROM ORDERS;

CUSTOMER

PRODUCT

CustomerID	Name	Email	Password	DateOfBirth
1	Shatakshi	shavi@gmail.com	shavi27	2007-02-27

PRODUCT

ProductID	ProductName	Description	Price	StockQty	Weight	SellerID	CategoryID
301	Laptop	GamingLaptop	75000.00	10	2.50	101	201

ORDERS

OrderID	OrderDate	OrderStatus	TotalAmount	CustomerID	AddressID
501	2026-08-29	Placed	75000.00	1	401

Output -3 :- Referential Integrity Violation

```
INSERT INTO ORDERS VALUES
(502, '2026-08-27', 'Placed', 50000, 999, 401);
```

Output

```
ERROR 1452 (23000):
Cannot add or update a child row:
a foreign key constraint fails
```

Reason

CustomerID = 999 does not exist in the CUSTOMER table.

Output 4:- ON DELETE SET NULL

```
SELECT ProductID, ProductName, CategoryID
FROM PRODUCT;
```

Before Deletion :-

ProductID	ProductName	CategoryID
301	Laptop	201

```
DELETE FROM CATEGORY
WHERE CategoryID = 201;

SELECT ProductID, ProductName, CategoryID
FROM PRODUCT;
```

After Deletion:-

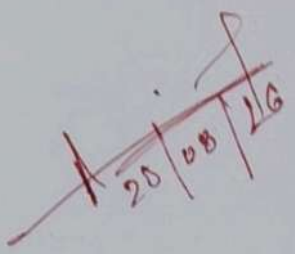
ProductID	ProductName	CategoryID
301	Laptop	NULL

Output -5:- ON DELETE CASCADE:-

```
SELECT * FROM ORDERS;
```

OrderID	OrderDate	OrderStatus	TotalAmount	CustomerID	AddressID
501	2026-08-29	Placed	75000.00	1	401

```
DELETE FROM CUSTOMER
WHERE CustomerID = 1;
SELECT * FROM ORDERS;
```



Output:-

Empty set

```
SELECT * FROM CUSTOMER_PHONE;
```

Output:-

Empty set